

Biography:

Gyu Rie Lee is an Assistant Professor in the Department of Biological Sciences at KAIST, where she leads research on AI-driven de novo protein design. She received her Ph.D. in Physical Chemistry from Seoul National University, developing computational methods for protein structure refinement and biomolecular interaction modeling and participating in the community-wide CASP structure prediction assessments. She subsequently joined Prof. David Baker's group at the University of Washington, first as a postdoctoral scholar and later as a staff scientist, where she worked until 2025. Her research integrates generative AI, computational modeling, and experimental validation to design functional proteins from scratch. Her lab focuses on generating proteins with novel molecular recognition and regulatory functions at atomic resolution, with applications in biomolecular sensing, regulatory switches, and the development of new-to-nature enzymes.

CURRICULUM VITAE

Gyu Rie Lee

이 규 리

gyurielee@kaist.ac.kr

WORK EXPERIENCE

- 2025.02 – Assistant professor, Dept. of Biological Sciences, KAIST
2023. 01 – 2025.01 Research Specialist I, University of Washington and Howard Hughes Medical Institute, Advisor: David Baker
2022. 07 – 2023. 01 Acting instructor, University of Washington, Advisor: David Baker
2018. 02 – 2022. 06 Postdoctoral scholar, University of Washington, Advisor: David Baker
2017. 03 – 2018. 01 Postdoctoral researcher, Seoul National University, Advisor: Chaok Seok

EDUCATION

2011. 09 – 2017. 02 Ph.D. in Physical Chemistry, Seoul National University, Advisor: Chaok Seok
Thesis: Development of protein modeling methods for structure refinement in the context of unreliable environments
2007. 03 – 2011. 08 B.S. in Chemistry Education, Seoul National University (*Cum Laude*)
Minor in Biology

PUBLICATIONS

- G. R. Lee, [...], D. Baker, De novo design of biotin-binding proteins, *manuscript in preparation*
- C. Lee, S. Choi, G. R. Lee, COMPASS: Decoupled latent steering for protein conformational transitions, ICML 2026 GenBio Workshop (2026).
- H. Seo, H. Ahn, [...], G. R. Lee, S. Ahn, J. Shin, S. Jung, VibeProteinBench: An evaluation benchmark for language-interfaced vibe protein design, arXiv:2605.10978 [q-bio.QM]
- M. Bauer*, J. Z. Zhang*, K. Wu*, G. R. Lee*+ (co-first and co-corresponding), B. Coventry*, [...], D. Baker+, De novo design of phospho-Tyrosine peptide binders, bioRxiv 2025.09.29.678898, *Nature Biotechnology*, *under revision*
- W. Yang*, S. Wang*, G. R. Lee*(co-first), J. Z. Zhang*, [...], D. Baker, The past, present and future of de novo protein design, *Nature* 652, 1139-1152 (2026).
- Z. Ghuo, O. Smutok, G. R. Lee, [...], E. Katz, D. Baker, and K. Alexandrov, Artificial allosteric protein switches with machine learning-designed receptors, *Nature Biotechnology* (2026).
- G. R. Lee*, S. J. Pellock*, C. Norn*, [...], A. H-W. Yeh, D. Baker, Small-molecule binding and sensing with a designed protein family, *Nature Communications* 17, 4533 (2026).
- E. Muratspahic, D. Feldman, D. E. Kim, [...], G. R. Lee, [...], D. Baker, De novo design of miniproteins targeting GPCRs, *Nature* (2026).
- H. Choi, [...], G. R. Lee, D. Hilvert, S. J. Pellock, D. Baker, Computational design of cysteine proteases, bioRxiv 2025.11.21.689808.
- L. Tran, [...], G. R. Lee, J. Mahamid, L. D. Lavis, L. An and D. Baker, Design of orthogonal far-red, orange and green fluorophore-binding proteins for multiplex imaging, bioRxiv 2025.08.03.668343. *Science*, *accepted* (2026).

- I. Anishchenko*, Y. Kipnis*, I. Kalvet*, [...], **G. R. Lee**, [...], D. Baker, Modeling protein-small molecule conformational ensembles with PLACER, *Proc. Natl. Acad. Sci. U.S.A.* 122 (45) e2427161122 (2025).
- J. Y.-H. Chen*, X. Peng*, C. Xi*, **G. R. Lee**, D. Baker, and A. H.-W. Yeh, De novo design of high-performance cortisol luminescent biosensors, *Journal of the American Chemical Society*, 147(31), 27494-27505 (2025).
- J. Dauparas, **G. R. Lee**, R. Pecoraro, L. An, [...], D. Baker, Atomic context-conditioned protein sequence design using LigandMPNN, *Nature Methods*, 22, 717-723 (2025).
- R. Krishna+, J. Wang, + W. Ahern+, P. Sturmfels, P. Venkatesh*, I. Kalvet*, **G. R. Lee*** (equal contribution, co-third), [...], D. Baker, Generalized biomolecular modeling and design with RoseTTAFold all-atom, *Science* 384, ead12528 (2024).
- L. An, M. Said*, L. Tran*, S. Majumder*, I. Goreshnik, **G. R. Lee**, [...], D. Baker, De novo design of diverse small molecule binders and sensors using Shape Complementary Pseudocycles, *Science*, 385, 276-282 (2024).
- C. J. Glasscock, R. Pecoraro, R. McHugh, [...], **G. R. Lee**, [...], D. Baker, Computational design of sequence-specific DNA-binding proteins, *Nature Structural and Molecular Biology*, <https://doi.org/10.1038/s41594-025-01669-4> (2025).
- S. Mansoor, M. Baek, H. Park, **G. R. Lee**, D. Baker, Protein Ensemble Generation through Variational Autoencoder Latent Space Sampling, *J. Chem. Theory Comput.* 20, 7, 2689–2695 (2024).
- S. Vázquez Torres, P. J. Y. Leung, I. D. Lutz, P. Venkatesh, J. L. Watson, F. Hink, H.-H. Huynh, A. H.-W. Yeh, D. Juergens, N. R. Bennett, A. N. Hoofnagle, E. Huang, M. MacCoss, M. Expòsit, **G. R. Lee**, P. M. Levine, X. Li, M. Lamb, E. N. Korkmaz, [...], D. Baker, De novo design of high-affinity protein binders to bioactive helical peptides, *Nature* 626(7998):435-442 (2024).
- H.-W. Yeh, C. Norn, Y. Kipnis, D. Tischer, S. J. Pellock, D. Evans, P. Ma, **G. R. Lee**, J. Z. Zhang, I. Anishchenko, [...], K. N. Houk & D. Baker, De novo design of high activity and specificity luciferases using deep learning, *Nature*, 614, 774–780 (2023).
- S. Lee, S. Kim, **G. R. Lee**, H. Woo, C. Seok & H. Park, Evaluating GPCR modeling and docking strategies in the era of deep learning-based protein structure prediction, *Computational and Structural Biotechnology Journal*, 21, 158-167 (2023).
- M. Baek, F. Dimaio, I. Anishchenko, J. Dauparas, S. Ovchinnikov, **G. R. Lee**, J. Wang, Q. Cong, L. N. Kinch, [...], D. Baker, Accurate prediction of protein structures and interactions using a three-track neural network, *Science*, 373(6557):871-876 (2021).
- L. An and **G. R. Lee**, De novo protein design using the blueprint builder in Rosetta, *Current Protocols in Protein Science*, 102(1):e116 (2020).
- H. Park, **G. R. Lee**, D. E. Kim, I. Anishchenko, Q. Cong, D. Baker, High-accuracy refinement using Rosetta in CASP13, *Proteins: Structure, Function, and Bioinformatics*, 87(12):1276-1282 (2019).
- **G. R. Lee**, J. Won, L. Heo, and C. Seok, GalaxyRefine2: Simultaneous refinement of inaccurate local regions and overall protein structure, *Nucleic Acids Res.*, 47, Web Server issue W451–W455 (2019).
- I. Bang, H. R. Kim, A. H. Beaven, J. Kim, S. Ko, **G. R. Lee**, H. Lee, W. Im, C. Seok, K. Y. Chung, H.-J. Choi, Biophysical and functional characterization of Norrin signaling through Frizzled4, *Proc. Natl. Acad. Sci.*, 115 (35), 8787-8792 (2018).
- C. Keasar et al. (**G. R. Lee**, C. Seok) An analysis and evaluation of the WeFold collaborative for protein structure prediction and its pipelines in CASP11 and CASP12, *Sci. Rep.*, 8:9939 (2018).
- J. Won, **G. R. Lee**, H. Park, and C. Seok, GalaxyGPCRloop: Template-Based and Ab Initio Structure Sampling of the Extracellular Loops of G-Protein-Coupled Receptors, *J. Chem. Inf. Model*, 58 (6), 1234-1243 (2018).
- **G. R. Lee**, L. Heo, and C. Seok, Simultaneous refinement of inaccurate local regions and overall structure in the CASP12 protein model refinement experiment, *Proteins: Structure, Function, and Bioinformatics*, 86 (S1), 168-176 (2018).
- X. Shao, H. Kang, T. Loveless T, **G. R. Lee**, C. Seok, W. I. Weis, H. J. Choi, and J. Hardin, Structural

- and functional characterization of the α -catenin- β -catenin binding interface in *Caenorhabditis elegans* reveals conserved requirements for cell-cell adhesion in metazoans, *J. Biol. Chem.*, in press (2017).
- Q. Xu, Q. Tang, P. Katsonis, O. Lichtarge, D. T. Jones, S. Bovo, G. Babbi, P. L. Martelli, R. Casadio, **G. R. Lee**, C. Seok, A. Fenton, R. L. Dunbrack, Jr., Benchmarking predictions of allostery in liver pyruvate kinase in CAGI4, *Human Mutation*, in press (2017).
 - H. Lee, M. Baek, **G. R. Lee**, S. Park, and C. Seok, Template-based modeling and ab initio refinement of protein oligomer structures using GALAXY in CAPRI Round 30, *Proteins: Structure, Function, and Bioinformatics*, 85, 399–407 (2017).
 - **G. R. Lee** and C. Seok, Galaxy7TM: Flexible GPCR-ligand docking by structure refinement, *Nucleic Acids Res.* 44 (W1), W502-506 (2016).
 - M. F. Lensik et al. (**G. R. Lee**, C. Seok), Prediction of homo- and hetero-protein complexes by protein docking and template-based modeling: a CASP-CAPRI experiment, *Proteins: Structure, Function, and Bioinformatics*, 84 (S1), 323-348 (2016).
 - W. -H. Shin, **G. R. Lee**, and C. Seok, Evaluation of GalaxyDock based on the Community Structure-Activity Resource 2013 and 2014 Benchmark Studies, *J. Chem. Inf. Model*, 56 (6), 988-995 (2016).
 - **G. R. Lee**, L. Heo, and C. Seok, Effective protein model structure refinement by loop modeling and overall relaxation, *Proteins: Structure, Function, and Bioinformatics*, 84 (S1), 293-301 (2016).
 - H. Park, **G. R. Lee** (co-first author), L. Heo, and C. Seok, Protein loop modeling using a new hybrid energy function and its application to modeling in inaccurate structural environments, *PLoS ONE* 9 (11): e113811 (2014).
 - W. -H. Shin, **G. R. Lee**, L. Heo, H. Lee, and C. Seok, Prediction of Protein Structure and Interaction by GALAXY protein modeling programs, *Bio Design*, 2 (1), 1-11 (2014).
 - **G. R. Lee**, W. Shin, H. Park, S. Shin, and C. Seok, Conformational sampling of flexible ligand-binding protein loops, *Bull. Korean Chem. Soc.* 33 (3), 770-774 (2012).

CONFERENCE TALKS

- 제 5회 한국생명정보학회 영종컨퍼런스, May 23rd, 2026.
- 2026년 한국물리학회 Spring meeting Tutorial session, April 22nd, 2026.
- 제 82 회 화학공학회 생물화공 심포지움: 글로벌 헬스 혁신을 위한 AI 기반 바이오/신약 발굴과 설계, Feb 23rd, 2026.
- 2026년 한국구조생물학회 PI 워크숍, Feb 5th, 2026.
- 2026 한국당과학회 동계워크숍, Jan 23th, 2026.
- 제13회 연세대학교 의과대학 유일한 의생명 포럼, Dec 17th, 2025.
- 2025년 KSMCB 에피유전체학분과 심포지움, Dec 10th, 2025.
- 서울대학교 국가미래전략원 주최 그랜드퀘스트 오픈포럼, Nov 27th, 2025.
- 2025 추계 한국공업화학회 AI계산공업화학분과회, Nov 7th, 2025.
- SMART Symposium: 2025 Frontier in Computational Chemistry, Biophysics, and Biological Sciences, Oct 25th, 2025.
- 한국생명정보학회 BIOINFO2025, Oct 16th, 2025.
- KSBB-AFOB Conference 2025, Sep 25th, 2025.
- 10th Polish-Korean conference on “Protein Folding: Theoretical and Experimental Approaches”, Sep 8th, 2025.
- 제17회 한국효소공학연구회 하계심포지움, Aug 22nd, 2025.

- 6th International conference for women in science and technology, Aug 20th, 2025.
- 제38회 농림식품산업 미래성장포럼 기조 연설, July 29th, 2025.
- 제1회 한국분자세포생물학회 미래컨퍼런스, July 16th, 2025.
- 대한화학회 생명화학분과 2025년 하계 워크샵, July 10th, 2025.
- 프레스티지 바이오메디컬 포럼: KAIST DAY, July 8th, 2025.
- 한국분자세포생물학회 리보핵산분과 2025년 하계 심포지엄, July 7th, 2025.
- 한국미생물생명공학회 2025 연례학술대회 및 국제 심포지엄, June 26th, 2025.
- 한국생명공학연구원 2025 단백질 디자인 심포지엄, Apr 4th, 2025.
- “Design of proteins interacting with chemical groups using AI”, ASK 2025, Jan 16th, 2025.
- “De novo design of small-molecule binding proteins using deep-learning”, UCSC Dept. of BME seminar, Jan 9th, 2025.
- “De novo design of small-molecule binding proteins using deep-learning”, AI Pharma Korea Conference 2024, Oct 31st, 2024
- “De novo design of small-molecule binding proteins using deep-learning”, 구조생물학회/KIDDS, July 5th, 2024
- “De novo design of small-molecule binding proteins using deep-learning”, KSBMB International Conference 2024, May 30th, 2024.
- “Towards generalizable design of de novo small-molecule binders using deep learning”, RosettaCon2023, August 9th, 2023.
- “Re-engineering ORs: Identification and Interrogation of Odorant Receptors in Bio-Hybrid Systems for Disease Detection”, Decoding Olfaction Convening organized by Bill & Melinda Gates foundation, March 9th, 2023.
- “Xe Sensing via Insights from Noble Gas-Protein Interactions”, 2019 DTRA Life Sciences Technical Review meeting, July 16, 2019.
- “Automatic protein structure refinement with an improved energy function and diverse sampling of unreliable regions”, CASP12 meeting, Refinement category round table, December 11, 2016.
- “Galaxy7TM: a web server for flexible GPCR-ligand docking”, The 118th Korean Chemical Society Meeting, October 13, 2016.
- “Predicting allosteric regulation using GALAXY protein modeling programs”, CAGI4 conference, March 25, 2016.

HONORS AND AWARDS

- KAIST Q-Day 특별표창 신뢰와 소통 부문 (2025. 11.25)
- Washington Research Foundation Innovation Postdoctoral Fellowship (2018. 02- 2020. 06)
- Best poster presentation award, Korean Society for Structural Biology (2017)
- Best poster presentation award, The 16th KIAS Conference on Protein Structure and Function, KIAS (2016)
- Scholarship for the Next Generation of Academics in Basic Research, Seoul National University (2015. 03 - 2017. 02)
- CASP11 Conference Participation Fellowship (financial supports of the participation in CASP11 meeting) (2014)
- Best poster presentation award, The 13th KIAS Conference on Protein Structure and Function, KIAS (2013)

- Best poster presentation award, 2013 Structure Festival, Korean Society for Structural Biology (2013)
- Best poster presentation award, The 12th KIAS Conference on Protein Structure and Function, KIAS (2012)
- Best presentation award, The 11th KIAS Protein Folding Winter School, KIAS (2012)
- Best poster presentation award, The 11th KIAS Conference on Protein Structure and Function, Korea Institute for Advanced Study (KIAS) (2011)